

A Survey of Encrypted Malicious Traffic Detection*

1st Yanmiao Li

Center of Information Security School of information science and engineering School of Computer Science and Technology
Beijing University of Posts and Telecommunications
Beijing, China
31893124@qq.com

1st Hao Guo

Shandong University
Qingdao, China
202012751@mail.sdu.edu.cn

3rd Jiangang Hou

Beijing Institute of Technology
Beijing, China
houjiangang@126.com

4th Zhen Zhang

Shandong Police College
Jinan, China
13405315678@163.com

5th Tongqing Jiang

Shandong Gentle Technology co., Ltd
Jinan, China
jiangtq@gentiesoft.com.cn

6th Zhi Liu*

School of information science and engineering
Shandong University
Qingdao, China
liuzhi@sdu.edu.cn

Abstract—With more and more encrypted traffic such as HTTPS, encrypted traffic protects not only normal traffic, but also malicious traffic. Identification of encrypted malicious traffic without decryption has become a research hotspot. Combined with deep learning, an important branch of machine learning, encrypted malicious traffic detection has achieved good results. This paper reviews the detection of encrypted malicious traffic in recent years. Firstly, we classify encrypted malicious traffic. Secondly, we sorts out the extraction characteristics of encrypted malicious traffic, the key and difficult problems we are facing at present. Then, with encrypted malicious traffic detection technology as the main line, we summarized the current detection model from the four core aspects of data collection, data processing, model training and evaluation improvement. Finally, we analyze the problems and point out future research directions.

Index Terms—HTTPS; Deep Learning; Encrypted Malicious Traffic; Feature Extraction; Detecting Techniques

I. INTRODUCTION

A. Background

The 21st century seems to belong to the computer information network. In today's society, information and communication technologies are developing rapidly. While people benefit from the convenience of Internet services, they often do not realize that their own information and other security issues are facing threats. These threats are usually composed of two parts: one is that a large number of applications collect users' information excessively, and once this information is leaked and maliciously used, the consequences will be severe; the other is that some lawbreakers collect and sell personal information illegally. The network information security problems caused by such incidents gradually make information protection the focus of social attention. Hence, encryption traffic arises at the historic moment. Gartner report[1] shows that more than 80% of traffic in 2019 is encrypted, and Hyper Text Transfer Protocol over Secure Socket Layer (HTTPS) traffic is commonly used in encrypted traffic. In 2019 alone,

more than 1 million encryption certificates are issued every day[2].

B. HTTPS

HTTPS is also known as Hyper Text Transfer Protocol over Secure Socket Layer. That is, the Secure Sockets Layer (SSL) layer is added under Hyper Text Transfer Protocol (HTTP), so the security base for HTTPS is Secure Sockets Layer/Transport Layer Security (SSL/TLS). It uses port 443 as the communication port by default, aiming to complete the communication that can only be encrypted and decrypted by client and server by using symmetric key and asymmetric key through Certificate Authority (CA) which is the public trust organization. The detailed handshake process is shown in Fig.1.

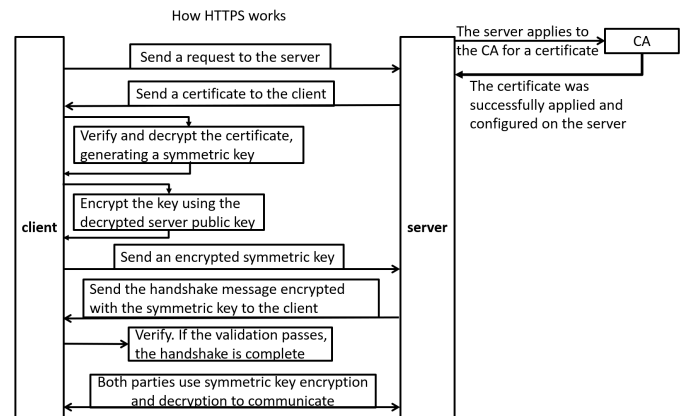


Fig. 1. How HTTPS works.

Firstly, the client sends a request to the server to connect to port 443 of the server, and the server sends the public key to the client with the CA certificate already applied. Secondly, the client's TLS verifies the public key and generates the symmetric key. The client then uses the public key to encrypt the symmetric key and sends it to the server, which

*Yanmiao Li and Hao Guo contributed equally to this work

*Corresponding author: Zhi Liu

decrypts it using the private key. Then the server sends symmetric key encrypted information to the client, and finally the client authenticates. If successful, the entire handshake is complete. The two sides conduct encryption and decryption communication[3]. This way ensures the security of both sides of communication.

C. Malicious traffic mode and its purpose

The transmission of such normal traffic is guaranteed by security, so encrypted transmission protects the privacy of users, as well as the malicious traffic. Malicious traffic can evade detection by transmitting network data in the form of encrypted traffic. That is, because the encryption techniques used to encrypt traffic are hard to crack, attackers can also use them to protect their information and hide their intentions. The methods in which they use encrypted traffic technology can be summarized into the following four categories, as shown in Table 1.

TABLE I
MALICIOUS TRAFFIC MODE AND ITS PURPOSE

mode	purpose
Encrypt normal file	Extort money from others
Encrypted malicious code	Bypass being killed
Encrypted network traffic	Avoid detection
Crack the code	Get permission

Cyren[4] found that nearly 40% of malware transmitted by encrypted traffic, and most of them transmitted by HTTPS. Therefore, it is urgent to detect the encrypted malicious traffic. However, it is a great challenge for network security defense to identify whether the encrypted traffic is malicious or not. Traditional traffic identification methods, such as deep packet detection and signature or pattern matching detection, can hardly help the identification of encrypted traffic. Therefore, it is an urgent matter to detect the encrypted malicious traffic. So we will describe the current situation of encrypted traffic detection in the next section.

D. Current status of encrypted malicious traffic detection

At present, the detection of encrypted malicious traffic mainly includes two ways according to the realization subject. One is to extract its features in a manual way for detection. The specific implementation principle is to analyze the implementation mode of malware and then extract its features for detection. This method is extremely labor-intensive, but relatively accurate and easy to implement. The other is an approach that has emerged in recent years, machine learning in artificial intelligence. Moreover, this method has made some achievements in image processing[5], speech translation[6], scene analysis[7], pattern recognition[8] and other fields, showing its superior performance. It can reduce human resources to some extent, and its accuracy can also be verified on the known data set. Detection methods can be divided into signature detection method and anomaly detection method[9]. The former is mainly to build a known detection model, and

the behavior or traffic with corresponding signatures will be judged as malicious traffic. The greatest advantage of this method is that it has a very high accuracy, but it can't detect unknown malicious traffic; The latter establishes rules based on normal behavior or traffic. Any irregular traffic is identified as malicious traffic. Although this method has certain defensive capabilities of the unknown flow, its biggest disadvantage is that it will lead to misjudgment, which will affect the user's experience. Therefore, this paper takes encrypted malicious traffic detection as the main line, and summarizes the members of malicious traffic, the research direction and achievements in recent years, as well as the subsequent difficult problems.

II. MALICIOUS TRAFFIC CLASSIFICATION

A. Malicious traffic classification

Scientific classification of malicious traffic is the key to extract malicious traffic characteristics so as to identify it, because different types of malicious traffic have different network behavior patterns. We need to select the corresponding feature extraction methods and identification methods according to the types of malicious traffic. However, due to the different research directions and different perspectives of researchers on malicious traffic, there is no uniform standard for the classification of malicious traffic. The JuMing classifies malicious traffic into 11 categories[10]. Anquanniu divided it into three main classes, among which there are several subclasses[11]. Yongan divides malicious traffic into collision library, cheating, malicious crawlers and other categories, among which the most are crawlers[12]. Rutkowska only classifies malicious traffic into four categories[13]. Malicious traffic is divided into four categories and 54 subcategories by Guancheng technology. Gang Lu and others classify malicious traffic into nine categories according to the attack effect of malicious traffic[14]. This paper classifies malicious traffic according to its usage and divides it into two categories: malicious software and malicious attack. The specific categories and corresponding descriptions are shown in Fig.2.

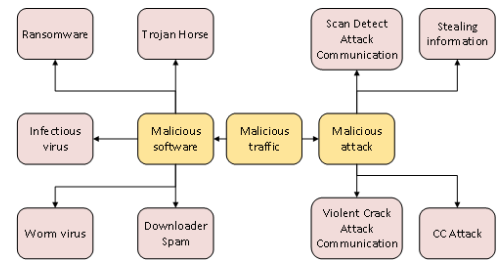


Fig. 2. Malicious traffic classification.

Challenge Collapsar (CC) attack[15] refers to the illegal elements using the proxy server to send a large number of seemingly legitimate requests to the affected server, causing the victim server to crash..

B. Variation of malicious family

In particular, it is important to emphasize that although there is an endless stream of malicious software, most of them are

variants of known malicious families. The Symantec study found that there were at least 357 million variants of malware in 2016 alone[16]. But it is obviously impossible to exhaust its variants. Fortunately, malicious encryption traffic detection is a dichotomous problem. Firstly, we are facing the problem of distinguishing malicious traffic from normal traffic, and then we can distinguish the families that malicious traffic belongs to, and turn the detection into a multi-classification problem to judge the family to which the malicious traffic belongs through the traffic characteristics, and this is a reversible problem. However, this must undergo a lot of training and analysis and discussion of the results to avoid the high false alarm rate.

III. MALICIOUS TRAFFIC FEATURE

A. Classification of features

Feature extraction, which relates to the detection accuracy and detection progress, is the most important part in the detection of encrypted malicious traffic. The selection of good features can help researchers analyze performance, shorten the analysis and groping time, and help to understand the implementation method and operation principle of encrypted malicious traffic detection. However, there is no unified standard and regulation for the characteristics of malicious traffic in the industry, so the classification of features is not unique. As discussed by Ferreira et al.[17], there is no clear uniform standard for analyzing network traffic except for the case of very specific traffic. Some people divide malicious traffic feature extraction into two categories, including purely statistical-based and content-based features. Others have classified feature engineering into four categories: spatiotemporal features, handshake features, background features and certificate features. Of course, the information collected, the extracted features and the selected algorithms are different for different research purposes. Traffic usually consists of 5 tuples[18], which are source IP, target IP, source port, target port and protocol. Generally speaking, traffic characteristics can be divided into three categories: content characteristics, data traffic statistical characteristics and network connection behavior characteristics[14]. The content characteristics of malicious traffic, which are also the most reliable features, mainly include the unique values of malicious traffic protocols and some very special characters and sequences contained in their payloads. Malicious data flow statistical characteristics and network connection behavior characteristics are the above-mentioned extraction methods based solely on statistical characteristics, which are described separately because there are still obvious differences between the two, but in fact, they are obtained from the statistical analysis of the collected data. Among them, the process of extracting statistical features of malicious data streams is generally to calculate the statistical values of data streams first, and then to extract malicious traffic characteristics based on these statistics, which is basically in the network layer, transport layer and application layer. In addition, network connection features are also generated by some malicious software, such as random scanning of IP, and generating a large number of failed connection worm

infections, etc. Different people have different classification methods for flow characteristics. Patel et al.[19] collected basic features, content features, 42 features based on time and host traffic. Meghdouri et al.[20] used the random forest method to analyze the performance of five feature sets, and it can also be divided into connection features, SSL features and certificate features. Villeneuve et al.[21] analyzed the content characteristics of Advanced Persistent Threat (APT) traffic. Lee et al.[22] proposed a method to automatically extract attack features, which can reduce the false positive rate very well.

B. TLS feature

As mentioned above, HTTPS can realize encrypted information transmission mainly because SSL/TLS is its security basis. Then when it comes to malicious traffic characteristics, we have to talk about the characteristics of TLS, which is generated on the basis of the SSL security protocol, and they are juxtaposed. Benign traffic has a unique TLS feature, which is absolutely impossible to be the same as malicious traffic. Therefore, given the universality and practicability of TLS protocol, TLS-based statistical feature extraction for encrypted malicious traffic has become a hot topic of current scholars' research. Anderson et al.[23] extracted the features of malicious TLS handshake traffic. TLS features mainly include the provided password group, the length of the client public key, TLS expansion and the password suite provided by the server, etc. These can be captured in requests sent to the server by the client when collecting traffic.

Blake Anderson et al.[23] found that the difference between normal traffic and malicious traffic on TLS is mainly in expansion and password suite. Most malicious traffic uses outdated password suite, such as 2048-bit RSA public key, while normal traffic uses 256-bit elliptic curve to encrypt public key. This provides a theoretical basis for correct classification.

C. Difficult problems of feature extraction

At present, the difficult problems of feature extraction of encrypted malicious traffic mainly focus on the following aspects:

First, as mentioned above, because there are many and complex variants of the malicious software family, which will lead to continuous changes in its network behavior, how to determine the common and accurate characteristics of malicious traffic is one of the difficulties of current research. Malicious encrypted traffic detection with incremental learning is the key to solve such problems.

Second, in the context of the rapid development of information technology, the network throughput is so high that it is difficult to record all the statistical information in the network, that is, if the detection speed does not keep up with the transmission speed of network data, then the data packets in it will be missed, resulting in unreported and affecting the system performance.

Third, the features that are being used such as TLS features, packet features and so on are indeed effective in detecting

encrypted malicious traffic, but they are far from enough in countering encrypted malicious traffic, because some such characteristics of malicious traffic such as data packets are very similar to normal traffic. Therefore, finding more effective malicious traffic characteristics is a difficult problem of great research value.

IV. ENCRYPTED MALICIOUS TRAFFIC DETECTION TECHNOLOGY

The essence of encrypted malicious traffic detection technology is information processing. Because of the popularity of encrypted traffic and the invisibility of load information, it is no longer remote to identify malicious encrypted traffic without decryption. Overcoming a large number of malicious traffic threats and challenges, and combining the advantages of reinforcement learning in artificial intelligence, a variety of machine learning-based detection methods have been proposed. Shahbaz Rezaei et al.[24] proposed a general framework for flow classification based on deep learning. They generalize the encrypted malicious traffic detection process into seven steps. This general model is already very representative and adapts to many algorithm models, but it does not cover unknown traffic, that is, classifiers fail to cover unrecognized traffic. Wang et al.[25] introduced an end-to-end encrypted traffic classification method model based on one-dimensional convolution neural network, which centralizes data processing into a unified end-to-end framework for cut processing, and then learns the characteristics by itself for classification. Derek Manning et al.[26] proposed an accelerated cryptographic traffic analysis system that utilizes an x86 multi-core platform and software accelerator configured at the edge of an enterprise network to minimize training and reasoning time while maintaining accuracy. Tangda Yu et al.[27] proposed an encrypted malicious traffic detection system based on multi-AEs (Autoencoder), which analyzed the characteristics of cryptographic protocols from handshake to authentication. For better classification, they also propose that the flow characteristics should be extracted further by extending the flow eigenvectors to higher positions, with higher accuracy. Wajdi Bazuhair et al.[28] uniquely proposed a new encoding method, which converts all the numerical data models into images when extracting features, and convolution neural networks are used as training and testing datasets after converting the connection features to images. It is worth mentioning that the data model carrier uses data enhancement methods, and uses Perlin noise as a carrier function to enhance the network traffic characteristic image to enhance the generalization of the depth model. The method of converting a numerical data model into an image is really novel, and the result of detection is sufficient. Incidentally, Danie et al.[29] concentrated on executable files, converted binary files into gray images, and classified binary data of malware samples using convolution neural network with 94.64% accuracy. It illustrates the role of convolution neural networks in detecting and classifying malicious software. Zhai Mingfang et al.[30] proposed a "six-step" model based on the seven-step detection method, which has a wider range

of applications and covers most of the studies, and it's a good combination of different detection models. Based on the existing research, this paper summarizes various models and methods, and summarizes the four core steps of encrypted malicious traffic. As shown in Fig.3.

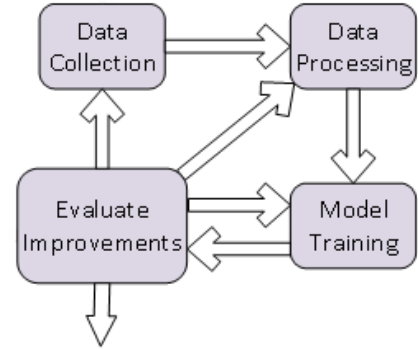


Fig. 3. Core steps for encrypted malicious traffic.

A. Data collection

If the problem is identified, the corresponding data will be collected to construct the available data set. This is also a key in encrypted malicious traffic detection. The best detection method also has the corresponding data set to reflect its value, otherwise it will have nothing to show. However, at present, most data set in the industry are internal, and can't be shared due to privacy and other issues, while the public data sets are mostly obsolete, which can't reflect the current trend of network traffic, or the public data set does not match the direction of their own research, so there is no truly representative data set. In addition, due to the evolution and variation of malicious software, its attack behavior is constantly changing, so the data set should be updated accordingly. Amirhossein Gharib[31] et al. defined a complete data set framework in view of the shortage of existing data sets. This framework contains 11 features, as shown in Table 2.

In a word, there is no effective public label data set and no universally acceptable labeling method in the current malicious encryption traffic detection technology[33]. At present, most research experiments use public data sets, and the existing public data sets include ISCX2012[34], ISCXVPN-nonVPN[35], DARPA data sets[36], KDDCUP99[37] and so on. These data sets are either collected directly, generated virtually, or combined as shown in Fig.4.

The data set collected directly can collect the real data of the real network, such as Wireshark packet capture. It reflects the actual situation of malicious traffic, but the disadvantage is that the proportion of malicious traffic in reality is too small, so that the marking is very difficult. The virtual generated data set can balance processing, strengthen the training process of deep learning, and eliminate the problem of small proportion of malicious traffic. However, human factors are relatively large, which can't accurately fit the real network. The combination of

TABLE II
DATA SET FRAMEWORK

Characteristic	describe
Complete Network configuration	It's actually a really world basis represented by an offline data set.
Complete Traffic	There can be real, pseudo-real or synthetic traffic in the data set.
Labeled dataset	Tagging and tagging data are very important.
Complete Interaction	Having all network interactions.
Complete Capture	In a complete traffic data set, all captured traffic has its corresponding research value, but due to equipment problems, there may be underreporting[32].
Available Protocols	A complete data set should have normal and abnormal traffic and its available protocols.
Attack Diversity	The types of attacks are changing and updating every day.
Anonymity	Privacy is violated when both IP and payload are available.
Heterogeneity	Datasets can be created with different data sources.
Feature set	The main purpose is to provide availability for researchers to test and analyze systems.
Metadata	The lack of appropriate metadata is one of the main problems with available data sets.

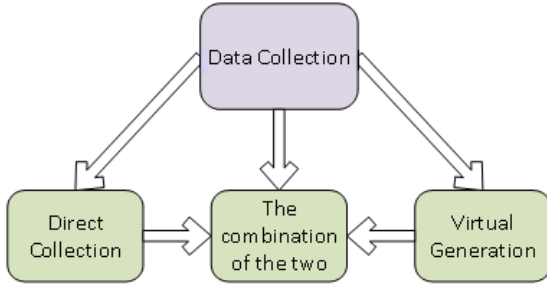


Fig. 4. Data collection method.

the two methods combines the advantages of the two methods, and can better meet the data requirements of malicious encrypted traffic detection, but this method does not significantly reduce the manpower requirements. Therefore, data collection is a basic but complex task. At present, the most important task is to construct a recognized tagging method and a well labeled and abundant high-quality data set.

B. Data processing

Data processing includes data preprocessing and feature extraction. Feature extraction has been introduced in detail in the previous chapter, so the following mainly describes data preprocessing. In order to ensure that the data set is complete and free of impurities, it is necessary to further process the original data set, that is, to conduct deep analysis and sorting, on this basis, to clean, integrate, mine, label and so on, so as to ensure the training quality of the later AI model, and make it become the data set meeting the requirements of deep learning test. Data discretization is also a very effective and efficient way to process[38]. Chen Tieming et al.[39] proposed a new method of sample enhancement based on key word

bank avoidance to process data for sample enhancement. Li Jia et al.[40] proposed a data preprocessing method based on semantic tailoring, adaptive field partition and statistical correlation. The idea is to transform the traffic data into matrix data, and then normalize the data into text information. Luo Ziming et al.[41] used the relief algorithm to preprocess the data to reduce the data dimension and improve the accuracy of the model. Relief algorithm is a feature weighting algorithm that gives different weights to features based on the correlation of each feature and category, and features with weights less than a certain threshold are removed. There are many more ways to preprocess data, so different preprocessing methods should be chosen according to the model.

C. Model training

The model is the main part of malicious encryption traffic detection technology, and it is also the most important part of traffic detection. At present, various algorithm models have been proposed to detect encrypted traffic in different situations. Shekhawat et al.[42] compared three machine learning algorithms, namely SVM, XGBoost and random forest, and used these three algorithms to train and test models for feature analysis. The result is that XGBoost performs slightly better than Random Forest, both with an accuracy of about 99%, while SVM performs relatively poorly. Baek et al.[43] proposed a kind of deep learning model based on which the average recall rate is 4.6 times of the existing model. Due to the maturity of engineering, different machine learning algorithms are used in different scenarios and then Guancheng Science and Technology[10] compared linear regression (LinReg), Random Forest (Random Forest), Decision Tree (DecTree), Neural Network (MLP), Support Vector Machine (SVM), Logical Regression (LogReg), Convolutional Neural Network (CNN), and finally found that random forest had the best comprehensive effect. Model selection is to select the appropriate model for the experimental goal in order to achieve the best detection results achieved by the existing technology.

If the model of neural network is used, a loss function needs to be defined when training the network as the direction of optimization parameters. At present, the neural network which studies the classification problem generally uses cross-entropy as the loss function[44]. After defining the loss function, each time the neural network is trained, the network parameters are updated by feedback after forward calculation. This process generally adopts back propagation algorithm (BP algorithm)[45]. In addition, it should be pointed out that over-fitting is a fundamental problem in deep learning, which refers to too much pursuit of the indicators of the training set, resulting in unsatisfactory performance in the test set. The main methods to reduce over-fitting are increasing training data, simplifying models, regularization, integrated learning methods and DropOut[46]. Researchers can choose appropriate methods to reduce over-fitting according to the actual situation. In order to avoid the contingency of the experiment, the N-fold cross-validation method is used in most experiments. The common method is ten-fold cross-validation,

that is, data are divided into ten parts, nine of which are fixed in turn as training data set, and the last one as validation set to verify the final training results.

D. Evaluate improvements

After the model training is completed, it is usually necessary to evaluate the model. However, this evaluation does not put the model into a real network environment to test its performance, because security and other costs are too high. Usually a certain evaluation index evaluation model is specified, including accuracy and real-time. In the classification algorithm, the following are several basic indicators[47]: true Positive case (TP) refers to the number of normal samples that are correctly identified, true negative case (TN) refers to the number of malicious samples that are correctly identified, false positive case (FP) refers to the number of normal samples that are misidentified and false negative case (FN) refers to the number of malicious samples that are misidentified, and their combination can represent the accuracy of the model. The formulas of various performance indicators are shown in [48].The precision formula is shown in formula 1.

$$\frac{TP}{TP + FP} \quad (1)$$

The formula of false positive rate is shown in formula 2.

$$\frac{FP}{FP + TN} \quad (2)$$

The recall rate formula is shown in formula 3.

$$\frac{TP}{TP + FN} \quad (3)$$

The accuracy formula is shown in formula 4.

$$\frac{TP + TN}{TP + TN + FP + FN} \quad (4)$$

Among them, the coordinate system consisting of true positive rate as vertical axis and false positive rate as horizontal axis is called ROC curve[49],as shown in Fig.5.

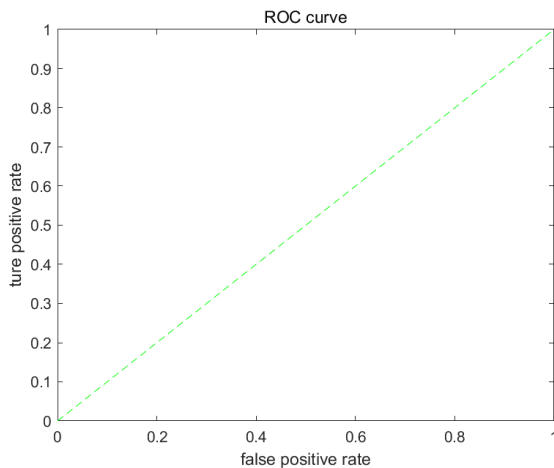


Fig. 5. ROC curve.

Points in the lower left corner of the coordinate graph represent cases in which a sample case is judged to be negative, while points in the upper right corner represent cases in which a normal case is judged, and dashed lines represent curves of random results. Ideally, the better classification should be in the upper left corner as far as possible, which means that the classifier has a high true positive rate while its false positive rate is low. The index to compare the performance of different ROC curves is to compare the area under the curve, the area of perfect classifier is 1, and the random classification is 0.5.

Real-time indicators are used to detect the efficiency of the model. Because malicious traffic has many variants and develops rapidly, real-time performance is used to evaluate the ability of encrypted traffic detection algorithms to keep pace with the times. At the same time, the models and algorithms are periodically updated to ensure their real-time performance. And continuously improve its detection ability.

V. SUMMARY AND PROSPECT

With the popularity of encrypted traffic, attackers also use encrypted traffic to realize their illegal intentions accordingly, which endangers the privacy security of network users. The importance of detection technology is becoming increasingly prominent. We should tell the illegal molecule in reality that the network is not an illegal place[50]. The detection of encrypted malicious traffic has made considerable progress, and its framework has been summarized by many researchers, so defending against malicious encryption traffic is no longer a fantasy. However, there are also some problems, or this is the main direction of the next research on malicious decryption traffic detection:

- Data sets: At present, most datasets are non-public datasets, which causes the experiment can't be reproduced, the same algorithm using different datasets results are biased, the different algorithms can't be compared among other issues, the real-time performance of public datasets is weak, and so on. Therefore, how to construct an accepted markup method to produce a relatively perfect and easy-to-implement dataset needs further study.
- Feature extraction of encrypted malicious traffic: Although the traditional statistical features of encrypted malicious traffic have achieved relatively good results, relying solely on these features is far from enough to combat encrypted malicious traffic. How to mine more effective and direct statistical features is still a difficult point in current research, which requires a lot of manpower and resources.
- applicability of encrypted traffic detection technology: Currently, researchers mainly focus on experimental algorithms and can't put them into the actual network for detection, because the real network has considerable uncertainty. So how to turn theory into reality is also a big problem.
- deep learning problems: As far as in-depth learning is concerned, some achievements have been made in the application of encrypted malicious traffic detection,

which has natural advantages and a variety of methods, so the possibilities are endless.

At present, defense only stays at the level of passive defense, that is, discovering attacks, finding measures, training algorithms to resist attacks. There is still a long way to go to convert passive defense into active defense.

REFERENCES

- [1] Cisco ETA - Provides Solution for Detecting Malware in Encrypted Traffic[EB/OL]. [2020/11/27]. <http://gbhackers.com/cisco-eta-encrypted-traffic/>.
- [2] Let's Encrypt Stats - Let's Encrypt - Free SSL/TLS Certificates[EB/OL]. [2020/11/27]. <https://letsencrypt.org/stats/>.
- [3] HTTPS Encryption mechanism - SXISZERO - Blog Garden [EB/OL]. [2020/11/27]. <https://www.cnblogs.com/sxiszero/p/11133747.html>.
- [4] SSL Traffic Growth - Malware is Moving Heavily to HTTPS - Cyren[EB/OL]. [2020/11/27]. <https://www.cyren.com/blog/articles/over-one-third-of-malware-uses-https>.
- [5] KRIZHEVSKY A, SUTSKEVER I, HINTON G E. ImageNet classification with deep convolutional neural networks[J]. Communications of the ACM, 2017,60(6): 84-90.
- [6] GRAVES A, MOHAMED A, HINTON G. SPEECH RECOGNITION WITH DEEP RECURRENT NEURAL NETWORKS[M]//International Conference on Acoustics Speech and Signal Processing ICASSP. 2013:6645-6649.
- [7] BYEON W, BREUEL T M, RAUE F, et al. Scene Labeling with LSTM Recurrent Neural Networks[M]//IEEE Conference on Computer Vision and Pattern Recognition. 2015:3547-3555.
- [8] FARIAS G, FABREGAS E, DORMIDO-CANTO S, et al. Automatic Recognition of Anomalous Patterns in Discharges by Applying Deep Learning[J]. FUSION SCIENCE AND TECHNOLOGY, 2020.
- [9] XIE Y, YU S. Anomaly detection based on Web users' browsing behaviors[J]. Journal of Software, 2007,18(4): 967-977.
- [10] Detection of Encrypted Communication Traffic of Malware - Know [EB/OL]. [2020/11/27]. <https://zhuanlan.zhihu.com/p/179245973>
- [11] A report on the first detection engine for encrypted traffic in China - Know [EB/OL]. [2020/11/27]. <https://zhuanlan.zhihu.com/p/59346026>.
- [12] Behind the "private domain flow": the interests promote the black industry chain, and malicious marketing is rampant - Know [EB/OL]. [2020/11/27]. <https://zhuanlan.zhihu.com/p/77692065>.
- [13] Introducing Stealth Malware Taxonomy[J].
- [14] LU Gang, GUO Ronghua, ZHOU Ying, WANG Jun. A Review of Malicious Traffic Feature Extraction [J].
- [15] LIU Jingyun. The principle and defense of CC attack [J]. Computer knowledge and technology(experience skill), 2016(12): 114-116.
- [16] SYMANTEC. 2017 Internet Security Threat Report[EB/OL]. [2020/11/27]. <https://www.symantec.com/content/dam/symantec/docs/reports/istr-22-2017-en.pdf>. 2017.
- [17] FERREIRA D C, VAZQUEZ F I, VORMAYR G, et al. A Meta-Analysis Approach for Feature Selection in Network Traffic Research[M]//2017:17-20.
- [18] OKADA Y, ATA S, NAKAMURA N, et al. Comparisons of Machine Learning Algorithms for Application Identification of Encrypted Traffic[M]//Xue-wen C, Dillon T, Ishbuchi H, et al. 2011:358-361.
- [19] PATEL S, SONDHI J. A Review of Intrusion Detection Technique using Various Technique of Machine Learning and Feature Optimization Technique[J].
- [20] MEGHDOURI F, ZSEBY T, IGLESIAS F. Analysis of Lightweight Feature Vectors for Attack Detection in Network Traffic[J]. APPLIED SCIENCES-BASEL, 2018,8(219611).
- [21] VILLENEUVE N, BENNETT J. detecting-apt-activity-with-network-traffic-analysis-whitepaper-en[J].
- [22] LEE S, KIM S, LEE S, et al. LARGen: Automatic Signature Generation for Malwares Using Latent Dirichlet Allocation[J]. IEEE TRANSACTIONS ON DEPENDABLE AND SECURE COMPUTING, 2018,15(5): 771-783.
- [23] BLAKE ANDERSON D M. Identifying Encrypted Malware Traffic with Contextual Flow Data[J].
- [24] REZAEI S, LIU X. Deep Learning for Encrypted Traffic Classification: An Overview[J]. IEEE Communications Magazine, 2019,57(5): 76-81.
- [25] WANG W, ZHU M, WANG J, et al. End-to-end encrypted traffic classification with one-dimensional convolution neural networks: IEEE, 2017.
- [26] MANNING D, LI P, WU X, et al. ACETA: Accelerating Encrypted Traffic Analytics on Network Edge[J].
- [27] YU T, ZOU F, LI L, et al. An Encrypted Malicious Traffic Detection System Based on Neural Network: IEEE, 2019.
- [28] BAZUHAIR W, LEE W. Detecting Malign Encrypted Network Traffic Using Perlin Noise and Convolutional Neural Network: IEEE, 2020.
- [29] GIBERT D, BEJAR J, MATEU C, et al. Convolutional Neural Networks for Classification of Malware Assembly Code[M]//Aguilo I, Alquezar R, Angulo C, et al. Frontiers in Artificial Intelligence and Applications. 2017:221-226.
- [30] ZHAI Mingfang, ZHANG Xingming, ZHAO Bo. Survey of encrypted malicious traffic detection based on deep learning [J]. Chinese Journal of Network and Information Security, 2020,6(3): 66-77.
- [31] GHARIB A, SHARAFALDIN I, LASHKARI A H, et al. An Evaluation Framework for Intrusion Detection Dataset[M]//2016:6.
- [32] RUAN Weiting, MA Zhaofeng. Research on Detection of Malicious Traffic in High-Speed Network: 2010 National Conference on communication security [C]. Kunming, Yunnan, China, 2010.
- [33] REZAEI S, LIU X. Deep Learning for Encrypted Traffic Classification: An Overview[J]. IEEE COMMUNICATIONS MAGAZINE, 2019,57(5): 76-81.
- [34] SHIRAVI A, SHIRAVI H, TAVALLAEI M, et al. Toward developing a systematic approach to generate benchmark datasets for intrusion detection[J]. COMPUTERS & SECURITY, 2012,31(3): 357-374.
- [35] DRAPER-GIL G, LASHKARI A H, MAMUN M S I, et al. Characterization of Encrypted and VPN Traffic using Time-related Features[M]//Camp O, Furnell S, Mori P. 2016:407-414.
- [36] 2000 DARPA Intrusion Detection Scenario Specific Datasets — MIT Lincoln Laboratory[EB/OL]. [2020/11/28]. <http://www.ll.mit.edu/r-d/datasets/2000-darpa-intrusion-detection-scenario-specific-datasets>
- [37] UCI KDD Archive[EB/OL]. [2020/11/28]. <http://kdd.ics.uci.edu/>.
- [38] WANG Jie, YANG Lili, YANG Min. Multitier ensemble classifiers for malicious network traffic detection [J]. Journal on Communications, 2018,39(10): 155-165.
- [39] CHEN Tieming, JIN Chengqiang, LYU Mingqi, ZHU Tiantian. Intelligent detection method on network malicious traffic based on sample enhancement [J]. Journal on Communications, 2020,41(06): 128-138.
- [40] LI Jia, YUN Xiaochun, LI Shuhao, ZHANG Yongzheng, XIE Jiang, FANG Fang. HTTP malicious traffic detection method based on hybrid structure deep neural network [J]. Journal on Communications, 2019,40(01): 24-33.
- [41] LUO Ziming, XU Shubin, LIU Xiaodong. Scheme for identifying malware traffic with TLS data based on machine learning [J]. Chinese Journal of Network and Information Security, 2020,6(1): 77-83.
- [42] SHEKHAWAT A S, TROIA F D, STAMP M. Feature analysis of encrypted malicious traffic[J]. Expert Systems with Applications, 2019,125: 130-141.
- [43] BAEK U, PARK J, HASANOVA H, et al. The Method of Seed Based Grouping Malicious Traffic by Deep-Learning: IEEE, 2018.
- [44] Detailed explanation of cross entropy loss function principle Cigar-CSDNBlog Cross entropy loss function [EB/OL]. [2020/11/28]. <https://blog.csdn.net/b1055077005/article/details/100152102>.
- [45] E R D, E H G, J W R. Learning representations by back-propagating errors[J].
- [46] WU Chunming, QIN Jian, XIE Deti. A learning algorithm for over fitting of BP network [J]. Journal of Southwest University (Natural Science Edition), 2008(01): 139-142.
- [47] Explanation of TP, TN, FP and FN Guyubit's blog CSDN blog [EB/OL]. [2020/11/28]. <https://blog.csdn.net/guyubit/article/details/52276013>.
- [48] ROC, Precision, Recall, TPR, FPR Understanding - Brief Book [EB/OL]. [2020/11/28]. <https://www.jianshu.com/p/be2e037900a1>.
- [49] ALONSO R, NAKAS C T, CARMEN PARDO M. A study of indices useful for the assessment of diagnostic markers in non-parametric ROC curve analysis[J]. COMMUNICATIONS IN STATISTICS-SIMULATION AND COMPUTATION, 2020,49(8): 2102-2113.
- [50] Network security level protection system 2.0 standard officially released—IT-People's Network [EB/OL]. [2020/11/28]. <http://it.people.com.cn/n1/2019/0516/c1009-31087714.html>.